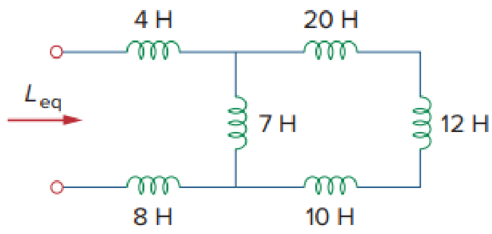
Chapter 6, Problem 11PP

Problem

Find the equivalent inductance of the circuit shown in Fig. 6.31.

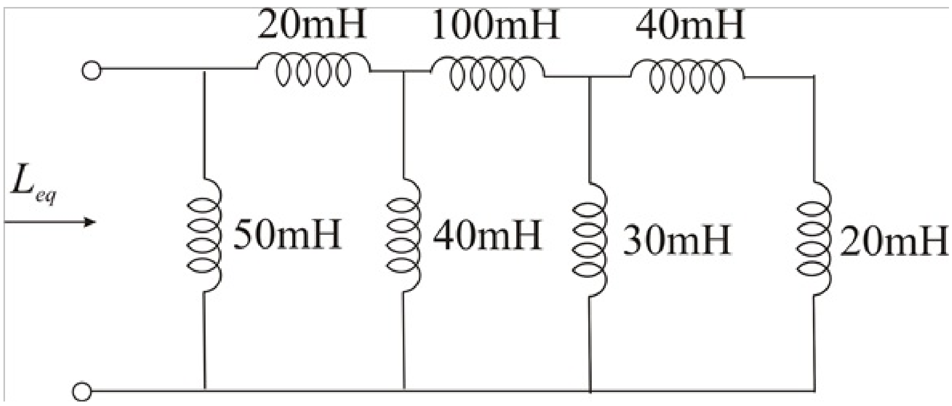
Fig. 6.31:



Step-by-step solution

Step 1 of 3

Given circuit



Step 2 of 3

In the above circuit 20mH and 40mH are in series

$$\left(20\times10^{-3}\right)+\left(40\times10^{-3}\right)=60\text{ mH}$$

This 60mH is in parallel with 30mH

$$60\text{ mH}\|30\text{ mH}=\frac{\left(60\times10^{-3}\right)\times\left(30\times10^{-3}\right)}{\left(60\times10^{-3}\right)+\left(30\times10^{-3}\right)}$$

$$60\text{ mH}\|30\text{ mH}=20\text{ mH}$$

Step 3 of 3

This 20mH inductor is in series with 100mH inductor,

$$\left(20\times10^{-3}\right)+\left(100\times10^{-3}\right)=120\text{ mH}$$

This 120mH inductor is parallel with 40mH inductor.

$$120\text{ mH}\|40\text{mH}=\frac{\left(120\times10^{-3}\right)\times\left(40\times10^{-3}\right)}{\left(120\times10^{-3}\right)+\left(40\times10^{-3}\right)}$$

$$120\text{ mH}\|40\text{ mH}=30\text{ mH}$$

This 30mH inductor is in series with 20mH inductor

$$\left(30\times10^{-3}\right)+\left(20\times10^{-3}\right)=50\text{ mH}$$

This 50mH inductor is in parallel with the 50mH inductor,

Hence

$$L_{eq}=\frac{\left(50\times10^{-3}\right)\times\left(50\times10^{-3}\right)}{\left(50\times10^{-3}\right)+\left(50\times10^{-3}\right)}$$

$$\therefore\boxed{L_{eq}=25\text{ mH}}$$